Do Indian Investors Also Follow the Pied Piper? A Case for Pre and Post Financial Crisis

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Abstract

Market-wide herding emerges when investors ignore firm specific information and follow the market. Following this simple approach, investors feel safe by imitating a larger group who may have some extraordinary information. The present paper explores the presence of herding behaviour in Indian Stock Market by implementing cross-sectional absolute deviation technique. The study also compares the presence of herding behaviour in per-crisis period and post-crisis period. The findings reveal strong evidence of herding during extreme market upturn while anti-herding dominated at times of extreme downturn. This ascertains that herding behaviour is asymmetric with respect to direction of market movement and dominates during extreme movements. Moreover, a change has been observed in the herding behaviour over time. During pre-crisis period, there was anti-herding for upward movement but herding has been observed for downward movement. It may be due to ambiguity created in the market during such reversals. Post-crisis period exhibits strong evidence for herding during upturns. Post-crisis herding behavior can be called "intentional", as it is based on market fundamentals rather than imitation. Positive outlook of investors towards Indian market lead to speedy recovery after financial crisis. Contrary to the popular belief, anti-herding dominated during period of financial crisis. (JEL Classification: G02; G01)

Keywords: Herding; Crisis; Efficiency; Return

Introduction

Behavioral finance is a realistic approach to model financial markets which promises to overcome limitations faced by classical methodology (Barberis, 2003). It provides accurate picture of markets by accepting that all investors are not rational. Irrational behaviour by agents may lead to deviation of stock price form its fundamental value. Efficient market hypothesis postulates that rational investors would identify such arbitrage opportunities and bring stock price back to its fair value in short time. On the other hand Barberis (2003) argues that arbitrage strategy in presence of irrational agents can be both risky and costly allowing mispricing to sustain for longer periods. Emergence of friction as mentioned above and presence of mass psychology makes sustained mispricing prominent in stock market, leading to transactions at inefficient prices (Christie and Huang, 1995). One of the causes of mass psychology that received attention in recent times is concept of "herding".

Banerjee (1992) defines herding as a case when investors start imitating other decisions makers rather than taking their own decisions. In such cases, investors feel safe being part of a larger group who may have some extraordinary information. It leads to inefficient price equilibrium; prices do not reflect all available information as agents joining the herd ignore the information they possess. Regulators of the stock market express concern over presence of herding that strikes at foundations of an efficient market and destabilizes the financial system (Bikhchandani and Sharma, 2001). Caparrelli et al. (2004) explain herding and its consequences with a beautiful instance. One night a man chose to go to restaurant A which had been suggested by an extremely trusted source. When he reached he observed that restaurant B was fully occupied with customers waiting outside while restaurant A had only few patrons. So he changed his decision and ate at restaurant B. The person simply discarded the information he possessed and followed the crowd assuming that other agents have extraordinary information. His belief in wisdom of crowd may backfire if we postulate that other customers behaved in similar fashion and the customer who reached restaurant B first had incorrect information. Fear of joining few customers made the man enter restaurant B and be part of the herd. Similar instances can be found in financial markets. After financial crisis "herd behaviour" became a derogatory term that symbolized market irrationality. The present paper delves into the possibility of presence of herd behaviour in Indian market. It also explores if herding behaviour is asymmetric with respect to direction of market movement. Further, it examines herding behaviour before and after financial crisis. The findings reveal presence of asymmetric herding behaviour in Indian market. We also notice drastic change in herding behavior before and after financial crisis.

The remainder of the paper is organised as follows. Section 2 provides review of literature and discussion on research questions. Section 3 covers detail about adopted measure of herding and discusses the methodology. Section 4 presents empirical evidence of herding in Indian stock market. Finally, section 5 discusses conclusion, limitations and future scope of research.

Literature Review and Motivation for the Study

In recent past, empirical research demonstrating presence or absence of herd behaviour has gained attention. This section reviews relevant contributions to existing literature on herding in stock market.

Shiller and Pound (1986) is the pioneer in the empirical research on herding behaviour. They conducted survey among institutional investors that showed institutional investors were influenced by decisions and information given by other professionals. Lakonishok et al. (1992) studied presence of herding and positive feedback trading among institutional investors. They found presence of herding behaviour and positive feedback trading for smaller stocks while there was little evidence of same for larger stocks. Grinblatt (1995) noticed absence of herding among mutual fund while Wermers et al. (1999) found the presence of herd behavior for small and growth oriented among mutual funds. Christie and Huang (1995) tested presence of herding for firms listed on NYSE. They found that US markets are somewhat efficient and herding is not a significant factor while predicting stock returns. Chang et al. (2000) extended the work done by Christie and Huang (1995). They examined herd behaviour in different international markets i.e. US, Hong Kong, Japan, South Korea and Taiwan. They found no evidence of herding in US and Hong Kong, partial evidence of herding in Japan and significant evidence of herding in emerging markets of South Korea and Taiwan. Lao and Singh (2011) and Economou et al. (2011) also confirm that herd behaviour is more likely to manifest in emerging markets.

Earlier research (McQueen et al., 1996; Chang et al., 2000; Caporale et al., 2008; Tan et al., 2008; and Lee et al., 2013) have provided evidence on presence of directional asymmetry i.e. relation of individual return dispersion as a function of market return depends on direction of market movement. They conclude that herding is more intense during downward movement of the market. Extant literature (Christie and Huang, 1995; Chang et al., 2000; Caparrelli et al., 2004; Lao and Singh, 2011; and Chiang et al., 2013) have also postulated that intense herding occurs during periods of extreme market returns in international markets. Both these results are extremely relevant with respect to recent financial crisis as extreme volatility exist in such periods (Schwert, 1989; Schwert, 1990; Koutmos, 1995).

With the growth of capital markets in India, research on herding has also attracted attention. Lao et al. (2011) and Poshakwale and Mandal (2014) found that herding behaviour exits in Indian market and is prevalent during large market movements. Laskshman et al. (2013)

concluded that herding is present in Indian market but it is not severe. Prosad et al. (2012) reported that Indian market is effiecient and herding is not severe.

However, there is lack of consensus among researchers on herding, which makes it one of the most debated topics in behavioural finance. Because of inconclusiveness about presence of herding in Indian market, the present paper re-examines the question "Does herding behaviour exist in Indian stock market?" In this paper we investigate if asymmetric relation exists between market-wide herding and direction of movement in Indian stock market. Another important objective of the present work is to find effect of extreme market returns on market-wide herding. Further, we examine if market-wide herding exacerbated Indian stock market movements during 2008 financial crisis. We have analysed pre-crisis and post-crisis sub-periods separately to study impact of financial crisis on herding behaviour.

Methodology and Data Collection

The methodology proposed by Chang et al. (2000) have been used in order to detect market-wide herding. This approach is centred on the idea that return dispersion, measured by cross-sectional absolute deviation (CSAD), will be low during periods of herding. It rests on the belief that market-wide herding would not allow individual stock returns to deviate from the mean market return.

The cross-sectional absolute deviation model (Chang et al., 2000) is based on return dispersion model (Christie and Huang, 1995). Christie and Huang (1995) opined that return dispersion can be a good indicator for presence of market-wide herding. Dispersion measures deviation of individual return from market return and increases when individual stock returns depart from market return. It is expected that the presence of market-wide herding would decrease dispersion, which is measured by cross-sectional standard deviation (CSSD) of return. They also suggest that market continuously moves between normal and extreme phases, and investors would make decision based only on market performance during large market movements. During normal phases investors behave rationally while they tend to follow masses during extreme movements. As a result during extreme conditions individual returns will form a cluster around market return, decreasing CSSD compared to normal market conditions. This is in contrast to rational asset pricing model where dispersion increases during large market movements because individual stocks have different sensitivities to market performance.

**Spurious herding occurs when investors take similar decision based on available information as opposed to intentional herding where investors imitate each other and ignore possessed information (Bikhchandani and Sharma, 2000).*

Christie and Huang (1995) estimated cross-sectional standard deviation of individual stocks using equation [1]. A low value of CSSD could be a sign of intentional herding or spurious herding*.

$$CSSD_t = \sqrt{\frac{\sum_{i=1}^{N}\left(R_{i,t-} R_{m,t}\right)^2}{N-1}} \qquad [1]$$

Where, $R_{i,t}$ is return of stock i at time t; $R_{m,t}$ is average return of sample; N is number of stocks in sample

The CSSD of returns obtained over time was regressed over extreme market phases (Christie and Huang, 1995). Dummy variables have been used to account for normal and extreme market phases, with $D^U = 1$ for returns in extreme right of return distribution (zero otherwise) and $D^L = 1$ for returns in extreme left of return distribution (zero otherwise). Market-wide herding is present if the coefficients of regression in equation [2] are negative and statistically significant. Estimated value of α will denote average dispersion of returns of the sample excluding time period corresponding to dummy variables.

$$CSSD = \alpha + \beta^L D_t^L + \beta^U D_t^U + e_t \qquad [2]$$

Where, D^L and D^U are dummy variables indicating extreme market phases

Christie and Huang (1995) measure of herding was affected by existence of outliers in the sample. Later on Chang et al. (2000) took input from Christie and Huang (1995) and provide a robust method to measure market-wide herding. Similar to Christie and Huang (1995), they assumed that rational asset pricing model would lead to higher dispersion during large market movement. In order to minimize effect of outliers they proposed use of cross-sectional absolute deviation (CSAD) as a measure of dispersion and performed similar regression (equation [3]) to detect presence of market-wide herding. A negative and significant value of β^L and β^U would indicate market-wide herding.

$$CSAD = \alpha + \beta^L D_t^L + \beta^U D_t^U + e_t \qquad [3]$$

Where, D^L and D^U are dummy variables indicating extreme market phases

With the aim of measuring herding over entire market return distribution rather than market return extremes as proposed by Christie and Huang (1995), Chang et al. (2000) specified following model:

$$CSAD = \alpha + \gamma_1 |R_{m,t}| + \gamma_2 (R_{m,t})^2 + e_t \qquad [4]$$

This regression depends on relationship between CSAD and market return in order to identify herd behavior. Standard asset pricing model assume that return dispersion is linearly related to market return. Absence of herding would result in positive value of γ_1 in equation [4]. However, when herding is present dispersion would decrease or increase significantly less in proportion to market return. Square of market return is added in model to capture such effect, and existence of non-linearity through negative value of γ_2 will indicate herding.

Chang et al. (2000) also checked if asymmetric relationship exists between herding and direction of market. They tested above hypothesis using regression in equation [5] and [6]. Similar to previous case, in absence of herding CSAD will be linearly related to market return but in presence of herding relation will be nonlinear. This non linearity will be captured by negative and statistically significant value of coefficient γ_2.

$$CSAD_t^{up} = \alpha + \gamma_1^{up} |R_{m,t}^{up}| + \gamma_2^{up} (R_{m,t}^{up})^2 + e_t \qquad [5]$$

$$CSAD_t^{down} = \alpha + \gamma_1^{down} |R_{m,t}^{down}| + \gamma_2^{down} (R_{m,t}^{down})^2 + e_t \qquad [6]$$

The dataset consists of daily returns for 1079 large liquid stocks* listed in National Stock Exchange (NSE), from January 2000 to December 2014. This data was collected from CMIE-Prowess database. As of 31st December 2014, selected firms represented a market capitalisation of 86525 Rs. Thousand Million. Among these are major companies such as TCS, Infosys, Wipro, RIL, SBI, ICICI, ITC, HUL and others. Set of Selected firms represent all major industries such as Drugs and Pharmaceuticals (59 firms), Computer Software (48 firms), Banking Services (40 firms), Infrastructural Construction (31 firms) and others.

**In CMIE-Prowess, 'Large liquid stocks' set is formed of the companies that satisfy all of the following conditions: (1) Companies having an average market capitalisation of Rs. 100 Crore in the last 365 days. (2) Companies that were traded for more than 80 per cent of the last 365 traded days. (3) Companies having a liquidity of more than 10 per cent, where liquidity is the ratio of the total traded volume of a company's stock on NSE for the past 365 days to its market capitalisation as on the last day of the 365 day period.*

Empirical Evidence in Indian Market

This section discuses regression results in context to research questions in hand. Table 1 reports descriptive statistics of daily cross-sectional absolute deviation (CSAD) of individual stocks and daily returns for equally weighted market portfolio. Aggregate data for 2000-2014 shows market return is highly volatile with the range of 23.74%. Such volatility during period of study is not surprising considering financial crisis of 2008. Minimum market return of -10.98% was recorded on 21st January 2008 and maximum return on 12.76% was recorded on 18th May 2009.

At first basic model proposed by Chang et al. (2000) is fitted to check for presence of herding in Indian stock market. Table 2 shows regression results corresponding to following equation:

$$CSAD = \alpha + \gamma_1 |R_{m,t}| + \gamma_2 (R_{m,t})^2 + e_t$$

In the regression, equally weighted return for large liquid stocks is considered as market return. Estimated value of γ_1 is 0.417, which implies that cross-sectional dispersion increases with increase in market return as suggested by asset pricing models. Model estimate γ_2 to be insignificant but slightly negative indicating linear relationship between CSAD and $|R_{m,t}|$.

According to this herding behaviour is not present in Indian stock market during the period of study. Further, we have checked for market-wide herding individually for each year; results show absence of such behaviour. During 2008, 2011, 2012 and 2013 estimated value of γ_2 is positive and significant, which is consistent with asset pricing model. It implies that individual stock returns diverges as from market return during extreme movements.

Table 1 – Descriptive statistics for daily CSAD and daily market return

Period	**Market Return (%)**					**CSAD**				
	Mean	Median	Max	Min	SD	Mean	Median	Max	Min	SD
2000-2014	**0.11**	**0.24**	**12.76**	**-10.98**	**1.54**	**2.19**	**2.03**	**7.74**	**0.94**	**0.72**
2000	-0.03	0.02	5.68	-5.51	1.60	3.24	3.08	6.28	1.57	0.93
2001	0.01	0.04	6.53	-6.84	1.66	2.66	2.53	7.72	1.44	0.74
2002	0.18	0.20	5.24	-4.53	1.26	2.08	1.97	5.02	0.97	0.64
2003	0.40	0.44	3.77	-3.60	1.28	2.25	2.21	4.21	0.94	0.61
2004	0.182	0.38	6.36	-10.04	1.82	2.15	2.07	5.14	1.15	0.55
2005	0.20	0.38	3.74	-6.31	1.28	1.89	1.82	2.96	1.11	0.37
2006	0.10	0.37	6.42	-8.13	1.70	2.07	1.95	6.00	1.19	0.57
2007	0.24	0.40	3.47	-5.39	1.32	2.13	2.02	3.72	1.48	0.44
2008	-0.39	-0.27	6.92	-10.98	2.51	2.76	2.56	7.74	1.43	0.90
2009	0.42	0.46	12.76	-7.10	2.04	2.56	2.44	7.00	1.45	0.72
2010	0.10	0.25	3.00	-4.84	1.16	1.69	1.64	3.09	1.16	0.30
2011	-0.16	-0.09	4.13	-4.42	1.26	1.75	1.66	3.38	1.08	0.37
2012	0.16	0.24	2.98	-3.37	0.93	1.71	1.63	2.79	1.03	0.39
2013	-0.01	0.11	2.66	-2.96	1.05	1.87	1.77	3.17	1.02	0.38
2014	0.26	0.34	5.85	-4.34	1.31	2.11	2.02	5.38	1.12	0.50

Table 2 – Regression results for the presence of herding behaviour

Period	Constant	$\lvert R_{m,t} \rvert$	$R_{m,t}^2$	Adjusted R^2
2000-2014	**1.737 (0.00)**	**0.417 (0.00)**	**-0.003 (0.275)**	**0.356**
2000	2.621 (0.00)	0.598 (0.00)	-0.480 (0.146)	0.203
2001	2.113 (0.00)	0.449 (0.00)	0.005 (0.697)	0.552
2002	1.0595 (0.00)	0.553 (0.00)	-0.0133 (0.561)	0.479
2003	1.739 (0.00)	0.531 (0.00)	-0.037 (0.351)	0.307
2004	1.723 (0.00)	0.349 (0.00)	-0.005 (0.282)	0.560
2005	1.656 (0.00)	0.250 (0.00)	-0.010 (0.449)	0.214
2006	1.740 (0.00)	0.256 (0.00)	0.012 (0.172)	0.497
2007	1.828 (0.00)	0.341 (0.00)	-0.023 (0.216)	0.259
2008	2.225 (0.00)	0.204 (0.00)	0.023 (0.003)	0.497
2009	2.087 (0.00)	0.317 (0.00)	-0.002 (0.784)	0.374
2010	1.489 (0.00)	0.029 (0.00)	0.008 (0.536)	0.380
2011	1.456 (0.00)	0.244 (0.00)	0.031 (0.056)	0.548
2012	1.481 (0.00)	0.233 (0.00)	0.059 (0.005)	0.512
2013	1.574 (0.00)	0.255 (0.00)	0.076 (0.030)	0.510
2014	1.808 (0.00)	0.255 (0.00	0.028 (0.097)	0.417

Numbers in parenthesis are p-values

Table 3 and Table 4 present regression results for different direction of market movements. Purpose behind performing regression is to check for asymmetric relationship between CSAD and market return. Table 3 reports regression results for the market state when equally weighted market return, $R_{m,t} > 0$. Regression for the complete time period shows γ_2^{up} slightly negative but insignificant, which can be considered as weak evidence for herding during market upswings. Estimated value of γ_2^{up} for individual years show strong evidence of anti-herding for 2001, 2006 and 2013. Table 4 contains regression results for market conditions when $R_{m,t} < 0$. Estimated value of γ_2^{down} for the whole time period insignificant. Regression on sub-periods give strong evidence for anti-herding for year 2008 and 2012

Table 3 – Regression results for herding behaviour during upward market movement

Period	Constant	$\lvert R_{m,t} \rvert$	$R_{m,t}^2$	Adjusted R^2
2000-2014	**1.687 (0.000)**	**0.513 (0.000)**	**-0.005 (0.199)**	**0.415**
2000	2.578 (0.000)	0.582 (0.000)	-0.034 (0.398)	0.251
2001	2.149 (0.000)	0.352 (0.000)	0.074 (0.000)	0.697
2002	1.625 (0.000)	0.664 (0.000)	-0.020 (0.499)	0.531
2003	1.806 (0.000)	0.574 (0.000)	-0.031 (0.536)	0.351
2004	1.833 (0.000)	0.241 (0.000)	0.036 (0.012)	0.547
2005	1.625 (0.000)	0.390 (0.000)	-0.045 (0.187)	0.241
2006	1.791 (0.000)	0.133 (0.000)	0.067 (0.000)	0.606
2007	1.927 (0.000)	0.124 (0.000)	0.096 (0.030)	0.030
2008	2.171 (0.000)	0.249 (0.000)	0.038 (0.039)	0.592
2009	2.027 (0.000)	0.410 (0.000)	-0.009 (0.178)	0.415
2010	1.542 (0.000)	0.066(0.000)	0.103 (0.011)	0.360
2011	1.477 (0.000)	0.221 (0.000)	0.036 (0.134)	0.499
2012	1.447 (0.000)	0.319 (0.000)	0.045 (0.167)	0.529
2013	1.588 (0.000)	0.174 (0.000)	0.170 (0.001)	0.587
2014	1.808 (0.000)	0.255 (0.000)	0.028 (0.097)	0.417

Numbers in parenthesis are p-values

Table 4 – Regression results for herding behaviour during downward market movement

Period	Constant	$\lvert R_{m,t} \rvert$	$R_{m,t}^2$	Adjusted R^2
2000-2014	**1.782 (0.00.)**	**0.301 (0.000)**	**0.006 (0.146)**	**0.320**
2000	2.664 (0.000)	0.626 (0.005)	-0.065 (0.220)	0.151
2001	2.111 (0.071)	0.419 (0.000)	-0.012 (0.426)	0.538
2002	1.586 (0.000)	0.284 (0.009)	0.030 (0.287)	0.494
2003	1.757 (0.000)	0.205 (0.189)	0.009 (0.863)	0.151
2004	1.663 (0.000)	0.282 (0.000)	0.000 (0.934)	0.717
2005	1.655 (0.000)	0.108 (0.183)	0.017 (0.296)	0.247
2006	1.784 (0.000)	0.179 (0.0531)	0.015 (0.256)	0.482
2007	1.850 (0.000)	0.149 (0.103)	0.007 (0.720)	0.269
2008	2.336 (0.000)	0.078 (0.314)	0.0326 (0.001)	0.487
2009	2.208 (0.000)	0.101 (0.290)	0.027 (0.125)	0.302
2010	1.504 (0.000)	0.125 (0.031)	0.023 (0.098)	0.521
2011	1.428 (0.000)	0.275 (0.001)	0.024 (0.288)	0.571
2012	1.534 (0.000)	0.091 (0.208)	0.087 (0.001)	0.529
2013	1.595 (0.000)	0.185 (0.137)	0.077 (0.116)	0.486
2014	1.757 (0.000)	0.298 (0.005)	-0.021 (0.439)	0.359

Numbers in parenthesis are p-values

To check for herding behaviour during extreme market movements dummy variables are introduced in regression equation indicating extreme market returns. Table 5 and Table 6 show regression results for extreme market movements during period of study. Estimate for γ_2^{up} for extreme 1% (right tail) of return distribution is negative 0.019 (significant at 5% level) indicating presence of herding for extreme positive returns. Table 6 shows regression results for extreme negative returns. Estimated value of γ_2^{down} for extreme return is positive and significant for all three thresholds indicating anti-herding during market stress situation.

Table 5 – Regression results for herding behaviour during extreme positive returns

Constant	$\lvert R_{m,t} \rvert * D_t^{up}$	$R_{m,t}^2 * D_t^{up}$	Adjusted R^2
Extreme up 1%			
2.188 (0.000)	0.546 (0.000)	-0.019 (0.017)	0.1251
Extreme up 5%			
2.135 (0.000)	0.428 (0.000)	-0.005 (0.284)	0.248
Extreme up 10%			
2.084 (0.000)	0.435 (0.000)	-0.005 (0.225)	0.335

Numbers in parenthesis are p-values

Table 6 – Regression results for herding behaviour during extreme negative returns

Constant	$\lvert R_{m,t} \rvert * D_t^{up}$	$R_{m,t}^2 * D_t^{up}$	Adjusted R^2
Extreme up 1%			
2.139 (0.000)	0.030 (0.802)	0.033 (0.024)	0.104
Extreme up 5%			
2.095 (0.000)	0.164 (0.000)	0.018 (0.006)	0.193
Extreme up 10%			
2.058 (0.000)	0.169 (0.000)	0.018 (0.000)	0.234

Numbers in parenthesis are p-values

Inspecting regression results for extreme directional market movements leads to conclusion that returns for individual stocks converge during extreme positive returns while diverge during extreme market negative returns. Divergence of individual stock returns during market fall may point towards presence of rational investors who leverage on arbitrage opportunities.

In Table 7 and Table 8, we check for presence of market-wide herding for three sub-periods: Pre-crisis (2000-2007), Crisis (2008) and Post-crisis. Two regressions were fitted based on direction of market movements to get better understanding of the phenomenon. Results show that during pre-crisis period there was anti-herding ($\gamma_2^{up} = 0.038$) for positive market movements and herding for negative market movements ($\gamma_2^{down} = -0.014$). Estimated values align with the view that investors seek consensus and imitate the market because of loss aversion leading to herding during short term reversals. For year 2008 γ_2^{up} and γ_2^{down} are positive and significant indicating anti-herding. We expected herding to be high during crisis, but results obtained points towards market rationality. For post-crisis period, herding is observed for upward movements rather than downward movements. It can be explained through presence of rational investors who grabbed arbitrage opportunity because of market undervaluation. Comparison of pre-crisis and post-crisis period is interesting as financial crisis triggered herding for upward market movements compared to herding during reversal before the crisis.

Table 7 – Regression results for herding behaviour before and after financial crisis

Period	State	Constant	$\lvert R_{m,t} \rvert$	$R_{m,t}^2$	Adjusted R^2
2000-2007	Pre-crisis	1.888 (0.000)	0.353 (0.000)	0.038 (0.000)	0.365
2008	Crisis	2.171 (0.000)	0.249 (0.000)	0.038 (0.039)	0.592
2009-2014	Post-crisis	-0.010 (0.000)	0.502 (0.000)	-0.010 (0.003)	0.474

Numbers in parenthesis are p-values

Table 8 – Regression Results for herding behaviour before and after financial crisis

Period	State	Constant	$\lvert R_{m,t} \rvert$	$R_{m,t}^2$	Adjusted R^2
2000-2007	Pre-crisis	1.853 (0.000)	0.385 (0.000)	-0.014 (0.051)	0.241
2008	Crisis	2.336 (0.000)	0.078 (0.314)	0.032 (0.001)	0.487
2009-2014	Post-crisis	1.609 (0.000)	0.245 (0.000)	0.016 (0.060)	0.391

Numbers in parenthesis are p-values

Concluding Remarks

The present paper discusses the presence of market-wide herding in Indian stock market during 2000-2014. It explores relationship of market-wide herding with direction of market movement, extreme market returns and financial crisis of 2008.

The analysis of empirical findings do not reveal any evidence of herding for the entire time period. The results indicate asymmetric relation of herding with direction of market movement. The results are not consistent with the theory of loss aversion, however, weak evidence of herding was found for upward movements. Presence of herding during extreme positive returns and anti-herding during extreme negative returns strengthens our finding on asymmetric nature of herding. Finally, there was no evidence of herding during financial crisis of 2008. In fact, anti-herding dominated as returns diverged during the crisis. It depicts that investors behaved rationally during time of financial crisis. Investment decisions were based on information available rather than market concensus. Further, examination of pre-crisis period observed presence of market-wide herding during short term reversals. Perhaps, it may be due to fear created in the market during such reversals. On the other hand, results for post-crisis period provide evidence for herding behaviour during upward movements. It shows positive outlook of investors towards Indian market after financial crisis of 2008.

The present research work is not free from certain limitations. First, CSAD approach applied in this paper measures herding with respect to market return. Such a measure does not provide insights on causes of herding. An interesting question would be to explore whether herding is due to individual investors or institutional investors. Other variables such as volume and volatility incorporated in the model would have given a more complete picture. Second, this research is limited to market-wide herding, industry wise analysis would add value and lead to better understanding of Indian market. Finally, the study is based on large liquid stocks in Indian market, empirically comparing results for large and small stocks would be of interest.

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